

Teaching Exam

Subject – Computer Science

Topic – DBMS



Lecture No. – 10

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Topics to be Covered

Topic



- 1:- Normalization ✓
- 2:- 1- NF (Normal Form) ✓
- 3:- Closer method to find candidate key ✓
- 4:- Some example related to closer method ✓
- 5:- Functional Dependency ✓
- 6:- Types of Functional Dependency ✓
- 7:- Trivial Function Dependency ✓
- 8:- Non-Trivial Functional Dependency ✓
- 9:- Semi Non-Trivial Function Dependency ✓
- 10- Multivalued Functional Dependency ✓
- 11- Transitive Functional Dependency ✓
- 12- Full and Partial Functional Dependency ✓
- 13- Conclusion. ✓



Topic : Closer Method

Option A- $\{CF\}^+ = \{C, F\}$

$= \{C, F, G\}$ (Using $C \rightarrow G$)

$= \{C, E, F, G\}$ (Using $F \rightarrow E$)

$= \{A, C, E, E, F\}$ (Using $G \rightarrow A$)

$= \{A, C, D, E, F, G\}$ (Using $AF \rightarrow D$)

Since, our obtained result set is same as the given result set, so, it means it is correctly given.

Option B- $\{BG\}^+ = \{B, G\}$

$= \{A, B, G\}$ (Using $G \rightarrow A$)

$= \{A, B, C, D, G\}$ (Using $AB \rightarrow CD$)

Since, our obtained result set is same as the given result set, so, it means it is correctly given.

Option C- $\{AF\}^+ = \{A, F\}$

$= \{A, D, F\}$ (Using $AF \rightarrow D$)

$= \{A, D, E, F\}$ (Using $F \rightarrow E$)

Since, our obtained result set is different from the given result set, so, it means it is not correctly given



Topic : Closer Method

Option D- $\{AB\}^+ = \{A, B\}$

$= \{A, B, C, D\}$ (Using $AB \rightarrow CD$)

$= \{A, B, C, D, G\}$ (Using $C \rightarrow G$)

X

Since, our obtained result set is different from the given result set, so, it means it is not correctly given.

Thus,

Option (C) and Option (D) are correct.

Question 3:- Find the closure and candidate key

Consider a relation $R (A, B, C, D, E, F, G)$ with the functional dependencies-

- $A \rightarrow BC$
- $\underline{BC} \rightarrow \underline{DE}$
- $D \rightarrow F$
- $CF \rightarrow G$



Topic : Closer Method

Closure of attribute D-

$$D^+ = \{ D \}$$

$$= \{ D, F \} \text{ (Using } D \rightarrow F \text{)}$$

We can not determine any other attribute using attributes D and F contained in the result set.

Thus,

$$\cancel{D^+} = \{ D, F \}$$

Note:- D attribute is not closer all other attribute so D attribute will not be candidate key.

Closure of attribute set {B, C}-

$$\{ B, C \}^+ = \{ B, C \}$$

$$= \{ B, C, D, E \} \quad \text{(Using } BC \rightarrow DE \text{)}$$

$$= \{ B, C, D, E, F \} \quad \text{(Using } D \rightarrow F \text{)}$$

$$= \{ B, C, D, E, F, G \} \quad \text{(Using } CF \rightarrow G \text{)}$$

Thus,

$$\{ B, C \}^+ = \{ B, C, D, E, F, G \}$$

NOTE:- Set of {B,C} Attribute is not closer to all other attribute so it will not be candidate key

$$\begin{aligned} D^+ &= \{ D, F \} \\ BC^+ &= \{ B, C, D, E, F, G \} \end{aligned}$$



Topic : Closer Method

Q4: Consider the relation scheme $R = \{E, F, G, H, I, J, K, L, M, N\}$ and the set of functional dependencies $\{ \{E, F\} \rightarrow \{G\}, \{F\} \rightarrow \{I, J\}, \{E, H\} \rightarrow \{K, L\}, K \rightarrow \{M\}, L \rightarrow \{N\} \}$ on R . What is the key for R ?

A. $\{E, F\}$

B. $\{E, F, H\}$

C. $\{E, F, H, \underline{K, L}\}$

D. $\{E\}$

~~$E \rightarrow E$~~

$\{E, F\} \rightarrow \{G\}$ (Candidate)

$\{F\} \rightarrow \{I, J\}$

$\{E, H\} \rightarrow \{K, L\}$ (Candidate)

$K \rightarrow M$

$L \rightarrow N$

$\{K, L\} = \{K, L, M\}$ (Candidate Key)

$\{E, F, H\}^+ = \{E, F, H, I, J\}$

$\{E, F, H, K, L\}^+ = \{E, F, H, K, L, M, N, G, I, J\}$

$\{E, F, H, K, L\}$ is a Super Key.

$\{E, F, H, K, L\}$ is a Minimal Super Key.

$\{E, F, H, K, L\}$ is the Primary Key.



Topic :Closer Method

Finding attribute closure of all given options, we get:

$$\{E, F\}^+ = \{EFGIJ\}$$

$$\{E, F, H\}^+ = \{EFHGIJLMN\}$$

$$\{E, F, H, K, L\}^+ = \{EFHGIJLMN\}$$

$$\{E\}^+ = \{E\}$$

$\{EFH\}^+$ and $\{EFHKL\}^+$ results in set of all attributes, but EFH is minimal. So it will be candidate key. So correct option is (B).



Topic : Closer Method

Q5->Consider a relation scheme $R = (A, B, C, D, E, H)$ on which the following functional dependencies hold: $\{A \rightarrow B, BC \rightarrow D, E \rightarrow C, D \rightarrow A\}$. What are the candidate keys of R ?

$BCH \rightarrow A, B, C, D, E, H$
 $A \rightarrow B$
 $BC \rightarrow D$

$\times \{AE, BE\}^+ = \{A, B, E, C, D\}$

$DEH \rightarrow D, E, H, A, C, B$
 $E \rightarrow C$
 $D \rightarrow A$

$\{AE, BE, DE\} = \{A, B, D, E, C, D\}$

(a) AE, BE

(b) AE, BE, DE

(c) AEH, BEH, BOH

(d) AEH, BEH, DEH

$\{AEH, BEH, BCH\} = \{A, B, C, E, H, D\}$

$\{AEH, BEH, DEH\}^+ = \{A, B, D, E, H, C\}$



Topic : Closer Method

$(AE)^+ = \{ABECD\}$ which is not set of all attributes. So AE is not a candidate key.
Hence option A and B are wrong.

$(AEH)^+ = \{ABCDEH\}$

$(BEH)^+ = \{BEHCDA\}$

$(BCH)^+ = \{BCHDA\}$ which is not set of all attributes. So BCH is not a candidate key. Hence option C is wrong.

So correct answer is D.



Topic : Functional Dependency

A functional dependency occurs when one attribute uniquely determines another attribute within a relation. It is a constraint that describes how attributes in a table relate to each other. If attribute A functionally determines attribute B we write this as the $A \rightarrow B$.

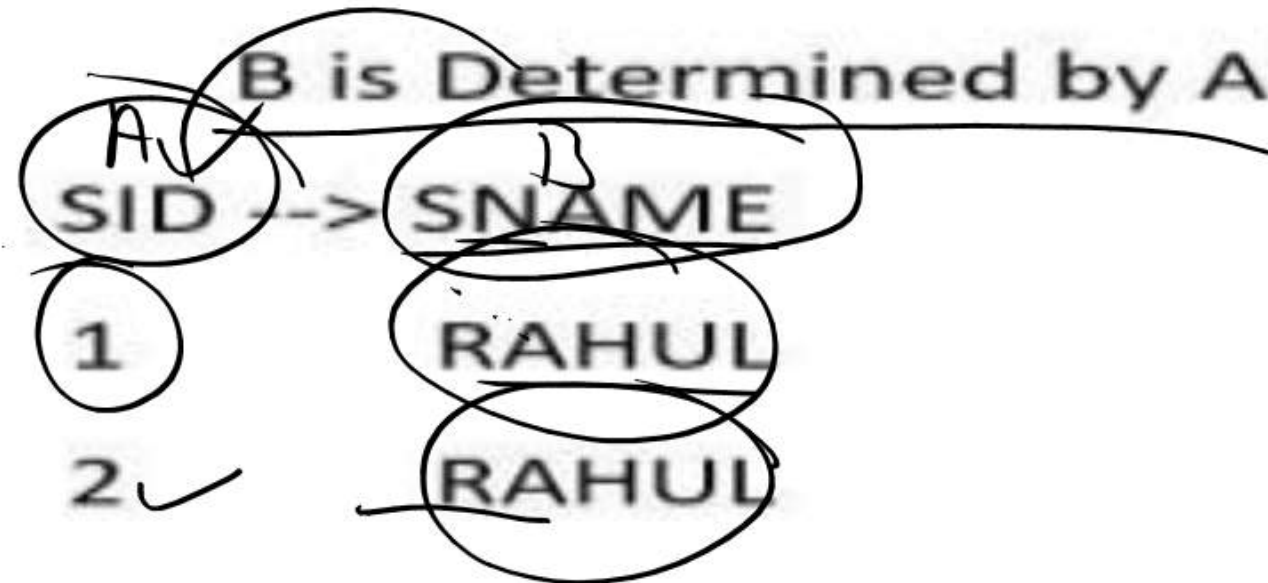
(Determinate) A $\rightarrow$ B (Dependent Attribute)

A Determinate B

OR

B is Determined by A

EX-





Topic : Functional Dependency

Functional dependencies are used to mathematically express relations among database entities and are very important to understanding advanced concepts in Relational Database Systems.

Example:

roll_no	name	dept_name	dept_building
42	abc	CO	A4
43	pqr	IT	A3
44	xyz	CO	A4
45	xyz	IT	A3
46	abc	EC	B2
47	jkl	ME	B2

Query
Select dept_name
from Company
where name = 'abc'

X {name} $\Rightarrow$ abc
 $\rightarrow$ {dept_name}



Topic : Functional Dependency

From the above table we can conclude some valid functional dependencies:

- $roll_no \rightarrow \{name, dept_name, dept_building\}$ → Here, roll_no can determine values of fields name, dept_name and dept_building, hence a valid Functional dependency
- $roll_no \rightarrow dept_name$, Since, roll_no can determine whole set of {name, dept_name, dept_building}, it can determine its subset dept_name also.
- $dept_name \rightarrow dept_building$, Dept_name can identify the dept_building accurately, since departments with different dept_name will also have a different dept_building
- More valid functional dependencies: $roll_no \rightarrow name$, $\{roll_no, name\} \rightarrow \{dept_name, dept_building\}$, etc.

Here are some invalid functional dependencies:

- $name \rightarrow dept_name$ Students with the same name can have different dept_name, hence this is not a valid functional dependency.
- $dept_building \rightarrow dept_name$ There can be multiple departments in the same building. Example, in the above table departments ME and EC are in the same building B2, hence $dept_building \rightarrow dept_name$ is an invalid functional dependency.
- More invalid functional dependencies: $name \rightarrow roll_no$, $\{name, dept_name\} \rightarrow roll_no$, $dept_building \rightarrow roll_no$, etc.



Topic :Types of Functional Dependency

Types of Functional Dependencies in DBMS

1. Trivial functional dependency
2. Non-Trivial functional dependency
3. Multivalued functional dependency
4. Transitive functional dependency

1. Trivial Functional Dependency

In Trivial Functional Dependency, a dependent is always a subset of the determinant. i.e. If $X \rightarrow Y$ and Y is the subset of X , then it is called trivial functional dependency.

Symbolically: $A \rightarrow B$ is trivial functional dependency if B is a subset of A .

The following dependencies are also trivial: $A \rightarrow A$ & $B \rightarrow B$

Example 1 :

- $ABC \rightarrow AB$
- $ABC \rightarrow A$
- $ABC \rightarrow ABC$



Topic : Trivial Functional Dependency

Example 1 :

- $ABC \rightarrow AB$
- $ABC \rightarrow A$
- $ABC \rightarrow ABC$

Example 2:

roll_no	name	age
42	abc	17
43	pqr	18
44	xyz	18

Here, $\{\text{roll_no}, \text{name}\} \rightarrow \text{name}$ is a trivial functional dependency, since the dependent name is a subset of determinant set $\{\text{roll_no}, \text{name}\}$. Similarly, $\text{roll_no} \rightarrow \text{roll_no}$ is also an example of trivial functional dependency.



Topic : Non-Trivial Functional Dependency



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2. Non-trivial Functional Dependency

In **Non-trivial functional dependency**, the dependent is strictly not a subset of the determinant. i.e. If $X \rightarrow Y$ and Y is not a subset of X , then it is called Non-trivial functional dependency.

Example 1 :

- $Id \rightarrow Name$
- $Name \rightarrow DOB$



Topic : Non-Trivial Functional Dependency

Example 2:

roll_no	name	age
42	abc	17
43	pqr	18
44	xyz	18

Here, $\text{roll_no} \rightarrow \text{name}$ is a non-trivial functional dependency, since the dependent name is not a subset of determinant roll_no. Similarly, $\{\text{roll_no}, \text{name}\} \rightarrow \text{age}$ is also a non-trivial functional dependency, since age is not a subset of $\{\text{roll_no}, \text{name}\}$



Topic :Semi NonTrivial Functional Dependency



3. Semi Non Trivial Functional Dependencies

A semi non-trivial functional dependency occurs when part of the dependent attribute (right-hand side) is included in the determinant (left-hand side), but not all of it. This is a middle ground between trivial and non-trivial functional dependencies. $X \rightarrow Y$ is called semi non-trivial when $X \cap Y$ is not NULL.



Topic :Semi NonTrivial Functional Dependency



3. Semi Non Trivial Functional Dependencies

A semi non-trivial functional dependency occurs when part of the dependent attribute (right-hand side) is included in the determinant (left-hand side), but not all of it. This is a middle ground between trivial and non-trivial functional dependencies. $X \rightarrow Y$ is called semi non-trivial when $X \cap Y$ is not NULL.



Topic :Semi NonTrivial Functional Dependency

Example:

Consider the following table:

Student_ID	Course_ID	Course_Name
101	CSE101	Computer Science
102	CSE102	Data Structures
103	CSE101	Computer Science

Functional Dependency:

$\{StudentID, CourseID\} \rightarrow CourseID$

This is semi non-trivial because:

- Part of the dependent attribute (*Course_ID*) is already included in the determinant ($\{Student_ID, Course_ID\}$).
- However, the dependency is not completely trivial because $\{StudentID\} \rightarrow CourseID$ is not implied directly.



Topic : Multivalued Functional Dependency



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4. Multivalued Functional Dependency

In Multivalued functional dependency, entities of the dependent set are not dependent on each other. i.e. If $a \twoheadrightarrow \{b, c\}$ and there exists no functional dependency between b and c , then it is called a multivalued functional dependency.

Example:

bike_model	manuf_year	color
tu1001	2007	Black
tu1001	2007	Red
tu2012	2008	Black
tu2012	2008	Red



Topic : Multivalued Functional Dependency



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4. Multivalued Functional Dependency

In Multivalued functional dependency, entities of the dependent set are not dependent on each other. i.e. If $a \twoheadrightarrow \{b, c\}$ and there exists no functional dependency between b and c , then it is called a multivalued functional dependency.

Example:

bike_model	manuf_year	color
tu1001	2007	Black
tu1001	2007	Red
tu2012	2008	Black
tu2012	2008	Red



Topic :Multivalued Functional Dependency



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In this table:

- X: bike_model
- Y: color
- Z: manuf_year

For each bike model (bike_model):

1. There is a group of colors (color) and a group of manufacturing years (manuf_year).
2. The colors do not depend on the manufacturing year, and the manufacturing year does not depend on the colors. They are independent.
3. The sets of color and manuf_year are linked only to bike_model.

That's what makes it a multivalued dependency.

In this case these two columns are said to be multivalued dependent on bike_model.

These dependencies can be represented like this:



Topic : Transitive Functional Dependency

In transitive functional dependency, dependent is indirectly dependent on determinant. i.e. If $a \rightarrow b$ & $b \rightarrow c$, then according to axiom of transitivity, $a \rightarrow c$. This is a transitive functional dependency.

Example:

enrol_no	name	dept	building_no
42	abc	CO	4
43	pqr	EC	2
44	xyz	IT	1
45	abc	EC	2

Here, $\text{enrol_no} \rightarrow \text{dept}$ and $\text{dept} \rightarrow \text{building_no}$. Hence, according to the axiom of transitivity, $\text{enrol_no} \rightarrow \text{building_no}$ is a valid functional dependency. This is an indirect functional dependency, hence called Transitive functional dependency.



Topic : Full and Partial Functional Dependency

6. Fully Functional Dependency

In full functional dependency an attribute or a set of attributes uniquely determines another attribute or set of attributes. If a relation R has attributes X, Y, Z with the dependencies $X \rightarrow Y$ and $X \rightarrow Z$ which states that those dependencies are fully functional.

7. Partial Functional Dependency

In partial functional dependency a non key attribute depends on a part of the composite key, rather than the whole key. If a relation R has attributes X, Y, Z where X and Y are the composite key and Z is non key attribute. Then $X \rightarrow Z$ is a partial functional dependency in RBDMS.



Topic :Conclusion on Functional Dependency



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Conclusion

Functional dependency is very important concept in database management system for ensuring the data consistency and accuracy.

THANK YOU